

Project Portfolio



Xyden – Leadership Maturity Assessment Platform | Xqtiv | Ongoing

Building an end-to-end adaptive leadership maturity assessment platform for senior executives, focusing on real-world judgment and trade-off decision-making rather than generic personality traits. The platform captures role context through time prioritization, assesses maturity via structured forced trade-off questions (100-point allocation), and scores participants into maturity levels (L1–L4) across Faculty, Behaviour, Technique, and Activity dimensions. Delivered the full UI end-to-end using an agentic automation pipeline, generating the initial working version within 1 month and completing all modules in 2 months — a project originally scoped at 6+ months.



Tech Stack: React, TypeScript, Tailwind CSS, Zustand

Key Features

- Adaptive leadership maturity scoring across L1–L4 levels
- Time Prioritization module to capture how leaders allocate attention
- Forced trade-off questions with 100-point allocation for nuanced assessment
- Report-ready dashboard and PDF generation using pre-authored narrative statements
- Five role-based modules: Candidate View, Test Admin, Client Manager, Content Admin, Super Admin
- End-to-end UI built using an agentic React app generation pipeline
- Currently in final testing and bug-fix phase ahead of production release



Impact: Delivered a full 6+ month project scope in just 2 months by leveraging an agentic UI generation pipeline, with all modules complete and nearing production release.

Recurring Transactions | Vanguard

Modernized a micro-frontend application for managing automatic investment and withdrawal plans. Worked closely with backend teams on API request/response design and seamless integration. Introduced an AI chatbot to assist clients with self-service tasks, reducing dependency on human support.

 **Tech Stack:** StencilJS, TypeScript, Jest, Agentic UI Builder

Key Features

- Micro-frontend architecture for modular, scalable development
- Automated investment and withdrawal plan management
- AI chatbot integration for client self-service
- Collaboration with backend teams on API design and integration
- Comprehensive unit testing with Jest for reliability

 **Impact:** Improved application performance and usability while reducing client support dependency through AI-driven self-service capabilities.

Agentic UI Builder | Vanguard

Contributed to the development of an agentic UI generation workflow that automates the creation of production-ready frontend components. The system leverages multiple structured inputs and follows a Test-Driven Development (TDD) approach, significantly accelerating the component development lifecycle.

 **Tech Stack:** React, Zustand, Shadcn UI, AWS Bedrock

Key Features

- Agentic workflow for automated UI component generation
- TDD-driven approach ensuring code quality from the ground up
- Integration with AWS Bedrock for AI-powered generation
- Shadcn UI for consistent, production-ready component output
- Structured input processing for precise component specifications

💡 **Impact:** Reduced component development time from 1-2 days to 1-2 hours, dramatically improving frontend team productivity.

🎨 Figma to JSON Converter | Vanguard

Developed a design extraction tool that bridges the gap between design and development by converting Figma designs into structured JSON with full component metadata. This tool enables automated UI generation pipelines, eliminating manual interpretation of design specifications.

🔧 **Tech Stack:** JavaScript, Tailwind CSS, HTML, CSS

Key Features

- Automated extraction of Figma design elements into structured JSON
- Full component metadata capture including styles, dimensions, and properties
- Seamless integration with automated UI generation pipelines
- Support for complex nested component structures
- Eliminates manual handoff between designers and developers

💡 **Impact:** Eliminated manual design interpretation by implementing an automated design-to-code pipeline, significantly reducing designer-to-developer handoff time.



Wealth Advisor Dashboard with Copilot | Capgemini

Engineered a comprehensive dual-mode financial dashboard with an integrated AI copilot chatbot. The platform serves both individual consumers and wealth advisors, offering rich financial insights, portfolio management tools, and personalized recommendations through a modern, responsive interface.



Tech Stack: React, Redux, Material UI, WebSocket

Key Features

- Dual-mode interface for consumers and wealth advisors
- AI copilot chatbot for real-time financial guidance
- Interactive portfolio analysis and risk assessment visualizations
- Financial goal management and progress tracking
- Appointment scheduling between clients and advisors
- Personalized article recommendations for investor education
- Real-time data updates via WebSocket connections



Impact: Enhanced investor decision-making and streamlined advisor consultations through personalized AI-driven insights and interactive visualizations.



Sustainability Chatbot | Synapse Green

Developed a user-friendly CSRD (Corporate Sustainability Reporting Directive) chatbot interface to simplify the understanding of reporting requirements for diverse companies. Also built an ESG reporting assistant featuring a mind map graph for enhanced brainstorming and a robust tabulation system for clear data presentation. Overcame technical challenges by implementing an innovative 'Read More' functionality for extensive text summaries.

 **Tech Stack:** React, Mind Map Graph, Data Tabulation

Key Features

- Interactive chatbot interface for CSRD reporting guidance
- Mind map graph visualization for ESG brainstorming and ideation
- Robust tabulation system for structured data presentation
- Innovative 'Read More' functionality for handling extensive text summaries
- Accessibility-focused design for diverse company types and users

 **Impact:** Significantly improved the efficiency and accuracy of complex ESG reporting processes while enhancing compliance accessibility for diverse organizations.

Octo Chatbot | Ongil

Developed a sophisticated chatbot application using React and WebSocket for seamless, long-lived connections. Integrated advanced chat components including interactive plots, dynamic tables, HTML summary displays, forms, file uploads, image displays, and PDF report generation. Implemented user-level customization to enable or disable specific functionalities.

 **Tech Stack:** React, Redux, Material UI, D3.js, WebSocket

Key Features

- Real-time WebSocket communication for seamless long-lived connections
- Interactive plots and dynamic tables for data-rich conversations
- HTML summary text display and form handling within chat
- File upload and image display capabilities
- Chat history management and export features
- PDF report generation directly from chat sessions

- User-level customization to toggle specific functionalities

💡 **Impact:** Enhanced both user experience and application flexibility through a highly extensible, feature-rich chatbot platform with personalized functionality control.

🎓 **Virtuosa Academic AI Assistant** | IIT Madras

Developed a personalized AI assistant designed to support students at IIT Madras with course-specific guidance and academic queries. The assistant provides targeted help by referencing specific video timestamps and lecture slides, enabling students to quickly find relevant content and deepen their understanding.

🔧 **Tech Stack:** React, Redux, Material UI

Key Features

- Personalized AI responses tailored to individual course content
- Precise references to specific video timestamps for targeted learning
- Slide-level references for accurate academic guidance
- Follow-up question suggestions to encourage deeper exploration
- Scalable architecture supporting 1,000+ concurrent student users

💡 **Impact:** Supported 1,000+ students in improving subject comprehension and academic performance through personalized, AI-driven academic assistance.

📊 **Data Analytics Dashboards** | 3M

Built a suite of high-performance data analytics dashboards for 3M, transforming complex raw datasets into rich, interactive visual insights. The dashboards empowered business stakeholders to make data-driven decisions with clarity and confidence through advanced

charting and visualization techniques.

 **Tech Stack:** React, Material UI, D3.js, Highcharts

Key Features

- Multiple high-performance dashboards tailored to business use cases
- Advanced interactive charts with drill-down and filtering capabilities
- D3.js custom visualizations for complex data representations
- Highcharts integration for rich, out-of-the-box charting
- Responsive design for cross-device compatibility

 **Impact:** Enabled data-driven decision-making by transforming raw enterprise data into meaningful, actionable visualizations for 3M business stakeholders.

Mentoring & Code Reviews | Ongil

Led structured mentoring and code review initiatives within the engineering team at Ongil. Established coding guidelines and best practices to elevate team-wide code quality, consistency, and collaboration. Played a key role in onboarding and training interns, accelerating their ramp-up and contribution to live projects.

 **Tech Stack:** Agile, Code Review, Technical Mentoring, Documentation

Key Features

- Structured intern training program with hands-on project guidance
- Regular code reviews to enforce quality and consistency standards
- Development of coding guidelines and best practice documentation
- Knowledge sharing sessions to upskill junior team members
- Fostered a collaborative engineering culture within the team

💡 **Impact:** Improved team code quality and accelerated intern onboarding by fostering a culture of best practices and continuous knowledge sharing.
